

GALILEO GALILEI



THE ITALIAN SCIENTIST Galileo Galilei was one of the greatest astronomers and physicists of all time. He was the first person to use a telescope to look at the heavens. He started a branch of physics called mechanics, showing that nature obeyed mathematical rules. His belief that science should be based on observation made him one of the first modern scientists. It also led him into trouble, because his views about the Solar System went against those held by the Roman Catholic Church.

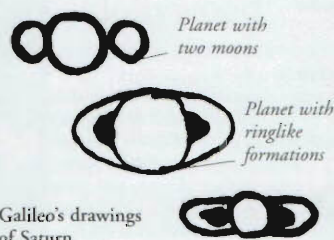
Telescope

In 1609, Galileo heard of the invention of the telescope and made one of his own. He used it to look at the heavens and made many astronomical discoveries. He noticed that the planet Venus has phases like the Moon. This gave support to the theory of Nicolaus Copernicus that the planets went round the Sun.



Replica of Galileo's telescope, 1609

Artist's impression of the Milky Way



Galileo's drawings of Saturn

Milky Way

In 1610, Galileo built a telescope that could magnify 1,000 times. It enabled him to see thousands of stars that no human being had ever seen before. He trained his new telescope on the Milky Way and found that it was a vast collection of stars, clustered together in groups of various sizes.

The Starry Messenger

In March 1610, Galileo published many of his discoveries in his book, *The Starry Messenger*. The book also showed that Copernicus was right to say that the Earth moved around the Sun, and that the Roman Catholic Church's idea of an unmoving Earth at the centre of the Universe was wrong. *The Starry Messenger* infuriated many churchmen.

Inquisition

Galileo's support for Copernicus's ideas outraged the Catholic Church because the priests thought that the Earth should be at the centre of the Universe. In 1633, the Church called Galileo to appear before its court, or inquisition, in Rome. The court ordered him to deny his beliefs under threat of torture. Galileo was forced to agree that Earth was the centre of the Universe, but was heard to mutter, "Yet it *does* move".



Trial of Galileo



Early life

Galileo was born in Pisa, Italy, in 1564. After school he went to the University of Pisa to study medicine. But Galileo was more interested in mathematics and physics, and left without a degree. By the time he was 25, he was back at the university – as professor of mathematics.

Moving bodies

In the 16th century, people believed that the Sun moved around Earth. Galileo did not agree with this, and developed Copernicus's theory of Earth moving around the Sun.



Falling feather

Coin falls at same speed

Gravity

Galileo showed that all objects fall at the same speed, no matter what their weight. Previously, people had believed that heavier objects fell faster. There is a story that Galileo proved his theory by dropping objects from the leaning tower of Pisa, but this is probably not true. He certainly did an experiment like this, in which objects of different weights were dropped in identical jars from which the air was pumped out.



Crater on Moon

Illustrated pages from *The Starry Messenger*, 1610

GALILEO GALILEI

- 1564 Born, Pisa, Italy.
- 1589 Becomes Professor of Mathematics at Pisa University.
- 1609 Makes his first telescope.
- 1610 Publishes *The Starry Messenger*.
- 1632 Publishes *Dialogue*, explaining the two theories of the universe.
- 1633 Sentenced by the Inquisition.
- 1642 Dies under house arrest at Arcetri, Italy.

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